

CHARISMA EDITIONS

BAC Application

API Technical Reference

Android Client · Node.js Backend · MongoDB Atlas

Request & Response Specification — v1.0

Document Type	Technical API Reference	Classification	Confidential — Internal Use
Version	1.0	Date	March 2026
Stack	Android / Retrofit 2 / Gson	Database	MongoDB Atlas (madaure cluster)

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1 Authentication — Get Device Token

Purpose — First call on application launch. Identifies the device and returns a signed or unsigned JWT token.

REQUEST

Method	GET
Endpoint	api/mobile/v1/token/{deviceId}
Auth	None — public endpoint
Path Param	deviceId : Android hardware device identifier

MONGODB DOCUMENT — Collection: users

```
{
  "_id"      : ObjectId("69875b5c24f26910688c70c0"),
  "deviceId" : "7a4bb9a10fe8546d",
  "firstName" : "zaki",
  "lastName"  : "zaki",
  "email"     : "zaki@gmail.com",
  "phone"     : "0558282926",
  "userType"  : 2,
  "_class"    : "org.madaure.backoffice.models.Client",
  "branches"  : [ "694704222d9c95cfedbab4fd" ]
}
```

RESPONSE — HTTP 200 OK

Unsigned device (DeviceToken):

```
{
  "deviceId" : "7a4bb9a10fe8546d",
  "signed"   : false
}
```

Signed device (FullToken):

```
{
  "deviceId" : "7a4bb9a10fe8546d",
  "signed"   : true,
  "clientId" : "69875b5c24f26910688c70c0",
  "contractId" : "69875b5c24f26910688c70c1",
  "soldKey"   : "09D9B3A9",
  "deleted"   : false
}
```

Response Content-Type: text/plain (raw JWT string, not JSON)

Field Mapping — Android Model

JWT Field	Java Class	Java Field	Persisted To (FILE_MANAGEMENT)
deviceId	DeviceToken	deviceId	SharedConstants.DEVICE
signed	DeviceToken	signed	(evaluated in memory — not persisted)

clientId	FullToken	clientId	SharedConstants.CLIENT_ID
contractId	FullToken	contractId	SharedConstants.CONTRACT_ID
soldKey	FullToken	soldKey	SharedConstants.SOLD_KEY
deleted	FullToken	deleted	(evaluated in memory — not persisted)

Note: signed == false → onNavigateToMainApp(false, -1) — Guest / registration flow.

Note: signed == true → onNavigateToMainApp(true, 2) + store clientId, contractId, soldKey, LAST_TIME_CONNECTION.

2 Branches — Get All Branches

Purpose — Retrieve all available educational branches. Used to populate the branch selector at registration and profile setup.

REQUEST

Method	GET
Endpoint	api/mobile/v1/branches
Auth	Authorization: Bearer {token}

MONGODB DOCUMENT — Collection: branches

```
{
  "_id"      : ObjectId("6946e205189a0fb7d66ba13d"),
  "name"     : "████████",
  "createdAt" : ISODate("2025-12-20T17:51:01.138Z"),
  "lastModifiedAt" : ISODate("2025-12-20T17:51:01.138Z"),
  "deleted"   : false,
  "user"     : "madaure"
}
```

RESPONSE — HTTP 200 OK

```
[
  { "id" : "6946e205189a0fb7d66ba13d", "name" : "████████" },
  { "id" : "694704222d9c95cfedbab4fd", "name" : "████ ██████████" }
]
```

Field Mapping — Android Model

JSON Field	Java Field (BranchResponse)	Type	Notes
id	id	String	MongoDB ObjectId as string
name	name	String	Branch display name (Arabic)

Note: Backend filters deleted == false before returning the list.

Note: Cached to SharedConstants.BRANCHES via FILE_MANAGEMENT.WRITE_JSON_OBJECT().

3 Subjects — Get Subjects by Branch ID

Purpose — Load all subjects available under a specific branch. Two Retrofit declarations target the same endpoint with different return types.

REQUEST

Method	GET
Endpoint	api/mobile/v1/subject/by-branch/{branchId}
Auth	Authorization: Bearer {token}
Path Param	branchId : MongoDB ObjectId string
Variant A	getSubjects() → Call – used in Student_Inscription_Fragment
Variant B	getSubjectsByBranchId() → Call> – used in Student_Bac_Profile

MONGODB DOCUMENT — Collection: subjects

```
{
  "_id"      : ObjectId("696e42676b6f94532275fe"),
  "branchId" : "694704222d9c95cfedbab4fd",
  "branch"   : "■■■■■ ■■■■■■■■",
  "userId"   : "693fd84937d0a3eb72c1c353",
  "user"     : "education",
  "name"     : "■■■■■ ■■■■■■■■",
  "createdAt" : ISODate("2026-01-19T14:40:39.675Z"),
  "updatedAt" : ISODate("2026-01-19T14:48:47.385Z"),
  "deleted"  : false
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "id"      : "696e42676b6f94532275fe",
    "name"    : "■■■■■ ■■■■■■■■",
    "branchId" : "694704222d9c95cfedbab4fd",
    "branchName" : "■■■■■ ■■■■■■■■"
  },
  {
    "id"      : "696e42776b6f94532275ff",
    "name"    : "■■■■■■■■■■",
    "branchId" : "694704222d9c95cfedbab4fd",
    "branchName" : "■■■■■ ■■■■■■■■"
  }
]
```

Field Mapping — Android Model

JSON Field	Java Field (SubjectDto)	Type	Notes
id	id	String	MongoDB ObjectId as string

name	name	String	Subject display name (Arabic)
branchId	branchId	String	Reference to parent branch
branchName	branchName	String	Denormalized from branches collection

Note: Cached to SharedConstants.SUBJECTS.

Note: Offline-first: cache displayed immediately, server refresh runs in background when connectivity is available.

4 Units — Get Units by Subject ID

Purpose — Retrieve all video units (introductory course blocks) for a subject. Displayed horizontally in the UI.

REQUEST

Method	GET
Endpoint	api/mobile/v1/units/{subjectId}
Auth	Authorization: Bearer {token}
Path Param	subjectId : MongoDB ObjectId string

MONGODB DOCUMENT — Collection: units

```
{
  "_id"      : ObjectId("696f7153fcf164c8c5e3cfbd"),
  "branchId" : "69470422d9c95cfedbab4fd",
  "branch"   : "■■■■ ■■■■■■",
  "subjectId" : "696e42fb676b6f9453227603",
  "subject"   : "■■■■■■ ■■■■■■■■■■",
  "createdBy" : "aymen",
  "userId"    : "695d1a17b66b96e2abebfee0",
  "name"      : "■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■■ ■■■■ ■■■■ ■■■■■■ ■■ ■■■■",
  "order"     : 1,
  "url"       :
  "https://firebasestorage.googleapis.com/v0/b/madaure-lap-final.appspot.com/...",
  "image"     :
  "https://firebasestorage.googleapis.com/v0/b/madaure-lap-final.appspot.com/...",
  "createdAt" : ISODate("2026-01-20T12:13:07.440Z"),
  "updatedAt" : ISODate("2026-02-18T15:53:26.296Z"),
  "_class"    : "org.madaure.backoffice.models.Unit"
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "id"      : "696f7153fcf164c8c5e3cfbd",
    "branch"   : "■■■■ ■■■■■■",
    "subject"  : "■■■■■■ ■■■■■■■■■■",
    "name"     : "■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■■...",
    "order"    : 1,
    "url"      : "https://firebasestorage.googleapis.com/...",
    "image"    : "https://firebasestorage.googleapis.com/..."
  }
]
```

Field Mapping — Android Model

JSON Field	Java Field (Unit)	Type	Notes
id	id	String	MongoDB ObjectId as string

branch	branch	String	Denormalized branch name
subject	subject	String	Denormalized subject name
name	name	String	Unit title
order	order	Integer	Display / sort order
url	url	String	Firebase Storage — video URL
image	image	String	Firebase Storage — thumbnail URL

Note: Mapped to VideoItem(id, title, subject, url, ", image) for UnitsAdapter (horizontal RecyclerView).

Note: Cached to SharedConstants.UNITS + ' ' + subjectId.

5 Lessons — Get Lessons by Subject ID

Purpose — Load all lessons for a given subject. Three endpoint variants exist for different sorting and access levels.

REQUEST

Method	GET
Endpoint	api/mobile/v1/lesson/{subjectId}
Variant 2	api/mobile/v1/lesson/unit/{subjectId} – sorted by createdAt DESC
Variant 3	api/mobile/v1/lesson/free/{subjectId} – free-tier lessons only
Auth	Authorization: Bearer {token}
Path Param	subjectId : MongoDB ObjectId string

MONGODB DOCUMENT — Collection: lessons

```
{
  "_id"      : ObjectId("6993383780a037fd2dda4481"),
  "branch"   : "■■■■■ ■■■■■■■■",
  "branchId" : "694704222d9c95cfedbab4fd",
  "subjectId" : "696e4267676b6f94532275fe",
  "subject"  : "■■■■■ ■■■■■■■■",
  "unitId"   : "6993375080a037fd2dda4479",
  "unit"     : "■■■■■ ■■■■■■■■",
  "name"     : "■■■■■ ■■■■■■■■",
  "order"    : 1,
  "description" : "",
  "quiz"     : 20,
  "remarque"  : "",
  "createdBy" : "aymen",
  "createdAt" : ISODate("2026-02-16T15:31:03.218Z"),
  "updatedAt" : ISODate("2026-02-16T15:31:03.218Z"),
  "_class"   : "org.madaure.backoffice.models.Lesson"
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "id"      : "6993383780a037fd2dda4481",
    "unit"    : "■■■■■ ■■■■■■■■",
    "branch"  : "■■■■■ ■■■■■■■■",
    "subject" : "■■■■■ ■■■■■■■■",
    "name"    : "■■■■■ ■■■■■■■■",
    "order"   : 1,
    "description" : "",
    "quiz"    : 20,
    "percentage" : null,
    "remarque" : "",
    "createdBy" : "aymen",
    "createdAt" : "2026-02-16T15:31:03.218+00:00",
  }
]
```

```
        "updatedAt"    : "2026-02-16T15:31:03.218+00:00"
    }
]
```

Field Mapping — Android Model

JSON Field	Java Field (Lesson)	Type	Notes
id	id	String	MongoDB ObjectId as string
unit	unit	String	Denormalized unit name
branch	branch	String	Denormalized branch name
subject	subject	String	Denormalized subject name
name	name	String	Lesson title
order	order	int	Display / sort order
description	description	String	Optional description text
quiz	quiz	int	Total number of quiz questions for this lesson
percentage	percentage	Long	Nullable — quiz completion percentage (server-computed)
remarque	remarque	String	Optional teacher annotation
createdBy	createdBy	String	Creator username
createdAt	createdAt	String	ISO 8601 timestamp
updatedAt	updatedAt	String	ISO 8601 timestamp

Note: percentage field uses dest.writeValue() / in.readValue() in Parcelable to correctly handle null Long across Intent boundaries.

Note: Cached to SharedConstants.LESSONS + '_' + subjectId.

Note: Offline-first: cache shown to user immediately; server refresh occurs in background if connectivity is available.

6 Lesson Parts — Get Parts by Lesson ID

Purpose — Load all video/content parts comprising a single lesson. Parts are sorted client-side by order field.

REQUEST

Method	GET
Endpoint	api/mobile/v1/lesson-part/by-lesson/{lessonId}
Variant 2	api/mobile/v1/lesson-part/by-subject/{subjectId} – all parts for an entire subject
Auth	Authorization: Bearer {token}
Path Param	lessonId : MongoDB ObjectId string

MONGODB DOCUMENT — Collection: lesson_parts

```
{
  "_id"      : ObjectId("6995bd86837f665393bb12cd"),
  "branch"   : "■■■■■ ■■■■■■■■",
  "branchId" : "694704222d9c95cfedbab4fd",
  "subjectId" : "696e4267676b6f94532275fe",
  "subject"  : "■■■■■ ■■■■■■■■",
  "unitId"   : "6993375080a037fd2dda4479",
  "unit"     : "■■■■■ ■■■■■■■■",
  "lessonId" : "6993383780a037fd2dda4481",
  "lesson"   : "■■■■■ ■■■■■■■■",
  "order"    : 1,
  "title"    : "■■■■ ■■■■■ ■■ ■■■ ■■■■■ ■■■■■■■■■■",
  "url"      : "https://vimeo.com/1166048429",
  "duration" : 10,
  "body"     :
  "https://firebasestorage.googleapis.com/v0/b/madaure-lap-final.appspot.com/...",
  "createdBy" : "aymen",
  "createdAt" : ISODate("2026-02-18T13:24:22.445Z"),
  "updatedAt" : ISODate("2026-02-18T16:23:43.811Z"),
  "_class"    : "org.madaure.backoffice.models.LessonPart"
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "id"      : "6995bd86837f665393bb12cd",
    "lessonId" : "6993383780a037fd2dda4481",
    "lesson"   : "■■■■■ ■■■■■■■■",
    "unitId"   : "6993375080a037fd2dda4479",
    "unit"     : "■■■■■ ■■■■■■■■",
    "subjectId" : "696e4267676b6f94532275fe",
    "subject"  : "■■■■■ ■■■■■■■■",
    "branchId" : "694704222d9c95cfedbab4fd",
    "branch"   : "■■■■ ■■■■■■■■",
    "order"    : 1,
    "title"    : "■■■■ ■■■■■ ■■ ■■■ ■■■■■ ■■■■■■■■■■",
  }
]
```

```

    "url"      : "https://vimeo.com/1166048429",
    "duration" : 10,
    "body"     : "https://firebasestorage.googleapis.com/...",
    "createdBy" : "aymen",
    "createdAt" : "2026-02-18T13:24:22.445+00:00",
    "updatedAt" : "2026-02-18T16:23:43.811+00:00"
  }
]

```

Field Mapping — Android Model

JSON Field	Java Field (LessonPart)	Type	Notes
id	id	String	MongoDB ObjectId as string
lessonId	lessonId	String	Reference to parent lesson
lesson	lesson	String	Denormalized lesson name
unitId	unitId	String	Reference to parent unit
order	order	int	Client-side sort key
title	title	String	Part title shown in UI
url	url	String	Vimeo video URL
duration	duration	int	Duration in minutes
body	body	String	Firebase Storage URL — PDF / supplementary notes

Note: Sorted client-side after fetch: `parts.sort((a,b) -> Integer.compare(a.getOrder(), b.getOrder()))`

Note: Cached to `SharedConstants.LESSON_PARTS + '_' + lessonId`.

7 Professors — Get Teachers by Subject / Branch

Purpose — Retrieve the teacher(s) assigned to a subject or branch. The best available active professor is selected client-side.

REQUEST

Method	GET
Endpoint	api/mobile/v1/prof/{subjectId}
Variant 2	api/mobile/v1/prof/branch/{branchId} – all teachers for a branch
Variant 3	api/mobile/v1/prof – all unrestricted teachers
Auth	Authorization: Bearer {token}
Path Param	subjectId or branchId : MongoDB ObjectId string

MONGODB DOCUMENT — Collection: profs

```
{
  "_id"          : ObjectId("6970d7643d12e16da863765d"),
  "branchId"     : "694704222d9c95cfedbab4fd",
  "branch"       : "■■■■■ ■■■■■■■■",
  "subjectId"    : "696e429f676b6f9453227601",
  "subject"      : "■■■■■ ■■■■■■■■■■■■",
  "fullName"     : "■■■■■ ■■■■ ■■■■■",
  "email"        : "ahmed.benali@example.com",
  "phone"        : "+213556123456",
  "image"        :
  "https://firebasestorage.googleapis.com/v0/b/madaure-lap-final.appspot.com/...",
  "experienceYear" : 12,
  "active"       : false,
  "level"        : "LYCEE",
  "description"   : "■■■■■ ■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■ ■■ ■■■■■■■■",
  "userId"       : "693fd84937d0a3eb72c1c353",
  "createdBy"    : "education",
  "createdAt"    : ISODate("2026-01-21T13:40:52.419Z"),
  "updatedAt"    : ISODate("2026-01-21T13:40:52.419Z"),
  "_class"       : "org.madaure.backoffice.models.Prof"
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "id"          : "6970d7643d12e16da863765d",
    "branch"       : "■■■■■ ■■■■■■■■",
    "subject"      : "■■■■■ ■■■■■■■■■■■■",
    "fullName"     : "■■■■■ ■■■■ ■■■■■",
    "email"        : "ahmed.benali@example.com",
    "phone"        : "+213556123456",
    "image"        : "https://firebasestorage.googleapis.com/...",
    "experienceYear" : 12,
```

```
{
  "active"      : false,
  "level"       : "LYCEE",
  "description" : "...",
  "createdAt"   : "2026-01-21T13:40:52.419+00:00",
  "updatedAt"   : "2026-01-21T13:40:52.419+00:00"
}
```

Field Mapping — Android Model

JSON Field	Java Field (Prof)	Type	Notes
id	id	String	MongoDB ObjectId as string
branch	branch	String	
subject	subject	String	
fullName	fullName	String	Displayed in teacher header UI
image	image	String	Loaded via Picasso
experienceYear	experienceYear	Integer	Nullable
active	active	boolean	Used in pickBestProf() selection logic
level	level	ELevel	Enum — e.g. LYCEE
description	description	String	

Note: Selection logic — pickBestProf(): (1) active==true AND fullName not empty, (2) fallback: any prof with fullName.

Note: Cached to 'PROF_BY_SUBJECT_' + subjectId.

Warning: experienceYear, active, level, and description are NOT written in writeToParcel(). These fields are lost if Prof is passed via Android Intent.

8 Client — Get & Update Student Profile

Purpose — Retrieve or update the authenticated student's profile. The GET response additionally injects contractId resolved from the contracts collection.

REQUEST

Method (GET)	GET
Endpoint	api/mobile/v1/client/{clientId}
Method (UPDATE)	PUT
Endpoint	api/mobile/v1/client/{clientId}
Auth	Authorization: Bearer {token}
Path Param	clientId : MongoDB ObjectId string
Body (PUT)	AddClient — { deviceId, firstName, lastName, email, phone, branchId }

MONGODB DOCUMENT — Collection: users

```
{
  "_id"      : ObjectId("69875b5c24f26910688c70c0"),
  "deviceId" : "7a4bb9a10fe8546d",
  "firstName" : "zaki",
  "lastName"  : "zaki",
  "email"     : "zaki@gmail.com",
  "phone"     : "0558282926",
  "userType"  : 2,
  "_class"    : "org.madaure.backoffice.models.Client",
  "branches"  : [ "694704222d9c95cfedbab4fd" ]
}
```

PUT request body (AddClient):

```
{
  "deviceId" : "7a4bb9a10fe8546d",
  "firstName" : "zaki",
  "lastName"  : "zaki",
  "email"     : "zaki@gmail.com",
  "phone"     : "0558282926",
  "branchId"  : "694704222d9c95cfedbab4fd"
}
```

RESPONSE — HTTP 200 OK (GET and PUT both return Client JSON)

```
{
  "contractId" : "69875b5c24f26910688c70c1",
  "id"          : "69875b5c24f26910688c70c0",
  "firstName"   : "zaki",
  "lastName"    : "zaki",
  "email"       : "zaki@gmail.com",
  "phone"       : "0558282926",
  "branches"    : [
    { "id" : "694704222d9c95cfedbab4fd", "name" : "■■■■■ ■■■■■■■■" }
  ]
}
```



```
    ],
    "userType"    : 2
  }
```

Field Mapping — Android Model

JSON Field	Java Field (Client)	Type	Notes
contractId	contractId	String	Resolved server-side from contracts collection
id	id	String	MongoDB ObjectId as string
firstName	firstName	String	
lastName	lastName	String	
email	email	String	
phone	phone	String	
branches	branches	List	Array of { id, name } objects
userType	userType	int	Always 2 for student accounts

Note: Cached to SharedConstants.CLIENT via FILE_MANAGEMENT.WRITE_JSON_OBJECT().

9 Key Validation — Validate Activation Key

Purpose — Verify that a sold token key is valid and unconsumed before proceeding with contract creation.

REQUEST

Method	POST
Endpoint	api/mobile/v1/key/{key}
Auth	Authorization: Bearer {token}
Path Param	key : alphanumeric activation key (e.g. 09D9B3A9)

MONGODB DOCUMENT — Collection: soldTokens

```
{
  "_id"      : ObjectId("6970bb6af66338689e48e9ee"),
  "userId"   : "693fd84937d0a3eb72c1c353",
  "createdBy" : "education",
  "key"      : "EDDF4C55",
  "code"     : "CH-M-E 260121",
  "createdAt" : ISODate("2026-01-21T11:41:30.351Z"),
  "updatedAt" : ISODate("2026-01-27T17:32:09.752Z"),
  "sold"     : true,
  "consumed" : false,
  "deleted"  : false,
  "_class"   : "org.madaure.backoffice.models.Key"
}
```

RESPONSE — HTTP 200 OK (plain text — not JSON)

```
"true"  — key is valid
          (sold == true AND consumed == false AND deleted == false)

"false" — key is invalid, already consumed, or marked deleted
```

Field Mapping — Android Model

Validation Condition	Field	Required Value	Application Action
Key has been sold	sold	true	Proceed to contract creation; persist key to SharedConstants.SOLD_KEY
Key not yet consumed	consumed	false	Key is available for use
Key not deleted	deleted	false	Key is active in the system

Note: On response 'true': key is written to SharedConstants.SOLD_KEY and the registration flow continues.

10 Contract — Create & Delete

Purpose — Register a new client and bind their account to an activation key. Deletion is OTP-verified and performs a soft delete.

REQUEST

Method (CREATE)	POST
Endpoint	api/mobile/v1/contract
Auth	Authorization: Bearer {token}
Body	AddContract – { client: AddClient, key: String }
Method (DELETE)	DELETE (Retrofit @HTTP annotation – DELETE with request body)
Endpoint	api/mobile/v1/contract
Body	RequestValidateOtp – { otp, clientId, contractId }

MONGODB DOCUMENT — Collection: contracts

```
{
  "_id"      : ObjectId("69875b5c24f26910688c70c1"),
  "clientId" : "69875b5c24f26910688c70c0",
  "expiringDate" : ISODate("2027-02-07T15:33:48.821Z"),
  "deleted"    : false,
  "createdAt"  : ISODate("2026-02-07T15:33:48.831Z"),
  "updatedAt"  : ISODate("2026-02-07T15:33:48.831Z"),
  "_class"     : "org.madaure.backoffice.models.Contract",
  "deviceId"   : "7a4bb9a10fe8546d",
  "soldKey"    : "09D9B3A9"
}
```

POST request body (AddContract):

```
{
  "client" : {
    "deviceId" : "7a4bb9a10fe8546d",
    "firstName" : "zaki",
    "lastName" : "zaki",
    "email" : "zaki@gmail.com",
    "phone" : "0558282926",
    "branchId" : "694704222d9c95cfedbab4fd"
  },
  "key" : "09D9B3A9"
}
```

RESPONSE — HTTP 200 OK

POST — raw signed JWT string (identical structure to endpoint 1 signed payload)
Application decodes the JWT and persists clientId, contractId, soldKey.

DELETE — void (empty response body)
Note: soldTokens.consumed is NOT reverted; the key remains consumed.

Field Mapping — Android Model

Side Effect	Collection	Operation	Notes
Create student account	users	Insert new document	Updated if deviceId already exists
Create contract record	contracts	Insert new document	Bound to clientId, soldKey, deviceId
Mark key as consumed	soldTokens	Set consumed = true	Prevents reuse of the activation key
Delete contract	contracts	Set deleted = true	Soft delete — triggered after OTP verification

11 Quiz — Get Question List & Submit Attempt

Purpose — Fetch all quiz questions for a lesson, then submit the student's answers. Two separate HTTP operations.

REQUEST

Method (GET)	GET
Endpoint	api/mobile/v1/quiz/{lessonId}/{clientId}
Auth	Authorization: Bearer {token}
Path Params	lessonId, clientId : MongoDB ObjectId strings
Method (SUBMIT)	POST
Endpoint	api/mobile/v1/quiz
Body	QuizAttemptRequest – { lessonId, clientId, submittedAnswers[] }

MONGODB DOCUMENT — Collection: quiz + quiz_attempt

quiz document:

```
{
  "_id"      : ObjectId("6978df3a53d578564e68d19e"),
  "lessonId" : "696f7a264a0fee59ea0e46dd",
  "question" : "■■■ ■■■■■■ ■■ ■■■■■■ ■■■■■■ ■",
  "options"  : [
    { "_id": "cc94b3d2-e52f-41a4-b0f0-deb2b4a95842", "content": "■■■■", "isCorrect": true },
    { "_id": "454e3166-881b-4ddc-a034-2f2867d67349", "content": "■■■■ ■■■", "isCorrect": false },
    { "_id": "5e016469-241c-46ac-87db-0bfda2da6837", "content": "■■■■", "isCorrect": false }
  ],
  "point"    : 10,
  "typeQuiz" : "SINGLESELECT",
  "createdBy" : "education",
  "_class"   : "org.madaure.backoffice.models.Quiz"
}
```

quiz_attempt document (created on POST):

```
{
  "_id"      : ObjectId("69945da0112fd0c53c16e5b2"),
  "lessonId" : "6993383780a037fd2dda4481",
  "clientId"  : "698e08094d29471542c8ffe6",
  "score"     : 3,
  "submittedAnswers" : [
    {
      "quizId"      : "69933a0a80a037fd2dda4493",
      "submittedAnswers" : [ "1854b964-0696-40ab-9539-b64941eff1c3" ],
      "score"       : 0
    }
  ],
  "createdAt" : ISODate("2026-02-17T12:22:56.679Z"),
  "updatedAt" : ISODate("2026-02-17T12:22:56.679Z"),
}
```

```

    "_class"      : "org.madaure.backoffice.models.QuizAttempt"
  }

  POST request body (QuizAttemptRequest):
  {
    "lessonId"      : "6993383780a037fd2dda4481",
    "clientId"      : "69875b5c24f26910688c70c0",
    "submittedAnswers" : [
      {
        "quizId"      : "6978df3a53d578564e68d19e",
        "submittedAnswers" : [ "cc94b3d2-e52f-41a4-b0f0-deb2b4a95842" ],
        "score"      : 1
      }
    ]
  }

```

RESPONSE — HTTP 200 OK

```

GET — list of Quiz questions:
[
  {
    "id"      : "6978df3a53d578564e68d19e",
    "lessonId" : "696f7a264a0fee59ea0e46dd",
    "question" : "■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■ ■",
    "options" : [
      { "id": "cc94b3d2-...", "content": "■ ■ ■ ■", "isCorrect": true },
      { "id": "454e3166-...", "content": "■ ■ ■ ■ ■ ■", "isCorrect": false },
      { "id": "5e016469-...", "content": "■ ■ ■ ■", "isCorrect": false }
    ],
    "point" : 10,
    "typeQuiz" : "SINGLESELECT"
  }
]

POST — Quiz result summary:
{
  "id"      : "6978df3a53d578564e68d19e",
  "lessonId" : "6993383780a037fd2dda4481",
  "score" : 3
}

```

Field Mapping — Android Model

JSON Field	Java Field (Quiz)	Type	Notes
id	id	String	MongoDB ObjectId as string
lessonId	lessonId	String	Reference to parent lesson
question	question	String	Question text (Arabic)
options	options	List	Each option: id + content + isCorrect
point	point	int	Points awarded for a correct answer
typeQuiz	typeQuiz	String	SINGLESELECT or MULTISELECT

12 Quiz History — Save & Retrieve Attempt History

Purpose — Persist quiz attempt history per client and subject. The bulk endpoint is designed for offline sync on connectivity restore.

REQUEST

Method (single)	POST
Endpoint	api/mobile/v1/quiz/history
Method (bulk)	POST
Endpoint	api/mobile/v1/quiz/history/bulk
Method (GET)	GET
Endpoint	api/mobile/v1/quiz/history/{clientId}/{subjectId}
Auth	Authorization: Bearer {token}
Body (POST)	QuizAttemptHistoryItem or List

MONGODB DOCUMENT — Collection: quiz_attempt

```
{
  "_id"           : ObjectId("69945da0112fd0c53c16e5b2"),
  "lessonId"      : "6993383780a037fd2dda4481",
  "clientId"      : "698e08094d29471542c8ffe6",
  "score"         : 3,
  "submittedAnswers" : [ ... ],
  "createdAt"     : ISODate("2026-02-17T12:22:56.679Z"),
  "updatedAt"     : ISODate("2026-02-17T12:22:56.679Z"),
  "_class"        : "org.madaure.backoffice.models.QuizAttempt"
}
```

RESPONSE — HTTP 200 OK

```
POST single / bulk — void (empty response body)

GET — list of history items:
[
  {
    "lessonId"   : "6993383780a037fd2dda4481",
    "subjectId"  : "696e4267676b6f94532275fe",
    "clientId"    : "69875b5c24f26910688c70c0",
    "score"       : 8,
    "totalPoints" : 10,
    "attemptDate" : "2026-02-17T12:22:56.679+00:00"
  }
]
```

Field Mapping — Android Model

JSON Field	Java Field	Type	Notes
------------	------------	------	-------

lessonId	lessonId	String	Reference to lesson
subjectId	subjectId	String	Reference to subject
clientId	clientId	String	Reference to student
score	score	int	Points earned in this attempt
totalPoints	totalPoints	int	Maximum achievable points
attemptDate	attemptDate	String	ISO 8601 timestamp

Note: Bulk endpoint is intended for offline sync: attempts are queued locally and flushed to the server when connectivity is restored.

13 Weekly Subject Progress — Aggregated Performance by Week

Purpose — Retrieve per-week quiz performance for a client/subject pair. Data is aggregated server-side from the quiz_attempt collection.

REQUEST

Method	GET
Endpoint	api/mobile/v1/quiz/progress/{clientId}/{subjectId}
Auth	Authorization: Bearer {token}
Path Params	clientId, subjectId : MongoDB ObjectId strings
Query Params	year (int) month (int)

MONGODB DOCUMENT — Collection: quiz_attempt (server-side MongoDB \$group aggregation)

```
{
  "_id"      : ObjectId("..."),
  "lessonId" : "6993383780a037fd2dda4481",
  "clientId"  : "698e08094d29471542c8ffe6",
  "score"     : 3,
  "createdAt" : ISODate("2026-02-17T12:22:56.679Z")
  // Server groups documents by ISO week within the requested year + month
  // and sums score / totalPoints per week group
}
```

RESPONSE — HTTP 200 OK

```
[
  {
    "week"      : 1,
    "subjectId" : "696e4267676b6f94532275fe",
    "totalScore" : 45,
    "totalPoints" : 100,
    "percentage" : 45
  },
  {
    "week"      : 2,
    "subjectId" : "696e4267676b6f94532275fe",
    "totalScore" : 78,
    "totalPoints" : 100,
    "percentage" : 78
  }
]
```

Field Mapping — Android Model

JSON Field	Java Field (WeeklySubjectProgressDto)	Type	Notes
week	week	int	ISO week number within the requested month

subjectId	subjectId	String	Reference to subject
totalScore	totalScore	int	Sum of all quiz scores for the week
totalPoints	totalPoints	int	Maximum achievable points for the week
percentage	percentage	int	$\text{totalScore} / \text{totalPoints} * 100$

14 OTP — Request One-Time Password

Purpose — Dispatch an OTP to the student's registered contact (SMS / email). Required as the first step of the contract deletion flow.

REQUEST

Method	POST
Endpoint	api/mobile/v1/otp
Auth	Authorization: Bearer {token}
Body	RequestOtp – { email: String, phone: String }

MONGODB DOCUMENT — Collection: users (read-only — OTP is generated and dispatched externally)

```
{
  "_id" : ObjectId("69875b5c24f26910688c70c0"),
  "email" : "zaki@gmail.com",
  "phone" : "0558282926"
  // OTP generated server-side and sent via external SMS / email service
  // OTP value is NOT persisted in MongoDB
}
```

RESPONSE — HTTP 200 OK (void — empty response body)

No response body.

The OTP is subsequently submitted via the DELETE /contract request body:

```
{
  "otp" : "123456",
  "clientId" : "69875b5c24f26910688c70c0",
  "contractId" : "69875b5c24f26910688c70c1"
}
```

Note: OTP is single-use and time-limited — validated server-side during the DELETE /contract call.

Note: The phone/email used for dispatch is resolved from the users collection using the clientId embedded in the JWT.

— Caching Strategy — Offline-First Architecture

All API data is persisted locally via **FILE_MANAGEMENT**. Every data-loading operation in the application follows the offline-first pattern described below:

01	Read from FILE_MANAGEMENT cache → present data to the user immediately (no blocking spinner).
02	Check network connectivity via ConnectivityManager + NET_CAPABILITY_VALIDATED.
03	If online → execute API request → overwrite local cache → refresh UI with server data.
04	If offline → retain cached data and continue without interruption.

Cached Data	FILE_MANAGEMENT Key	Format
JWT Token	SharedConstants.TOKEN	Raw string
Device ID	SharedConstants.DEVICE	Raw string
Client ID	SharedConstants.CLIENT_ID	Raw string
Contract ID	SharedConstants.CONTRACT_ID	Raw string
Sold Key	SharedConstants.SOLD_KEY	Raw string
Client object	SharedConstants.CLIENT	JSON object
Branches list	SharedConstants.BRANCHES	JSON array
Subjects list	SharedConstants.SUBJECTS	JSON array
Units per subject	SharedConstants.UNITS + '_' + subjectId	JSON array
Lessons per subject	SharedConstants.LESSONS + '_' + subjectId	JSON array
Lesson parts	SharedConstants.LESSON_PARTS + '_' + lessonId	JSON array
Prof per subject	'PROF_BY_SUBJECT_' + subjectId	JSON object
Selected branch name	SharedConstants.SELECTED_BRANCH_NAME	Raw string
Selected branch ID	SharedConstants.SELECTED_BRANCH_ID	Raw string
Last connection time	'LAST_TIME_CONNECTION.time'	Long (epoch millis)
Sound preferences	'effect.sound' / 'text.sound'	Raw string

— MongoDB Collection Reference

The following table provides a complete index of all MongoDB collections in the **madaure** cluster and their corresponding Android model classes.

Collection	Key Fields	Android Model
users	deviceId, firstName, lastName, email, branches[]	Client, AddClient
branches	name, deleted	BranchResponse
subjects	branchId, name, deleted	SubjectDto
units	subjectId, name, order, url, image	Unit → VideoItem
lessons	subjectId, unitId, name, order, quiz	Lesson
lesson_parts	lessonId, order, title, url, body, duration	LessonPart
profs	subjectId, fullName, active, level, image	Prof
quiz	lessonId, question, options[], point, typeQuiz	Quiz
quiz_attempt	lessonId, clientId, score, submittedAnswers[]	QuizAttemptRequest
contracts	clientId, expiringDate, deviceId, soldKey, deleted	AddContract
soldTokens	key, code, sold, consumed, deleted	(key validation)
progression	clientId, unitId, lessonId, progressionsPart[]	(not yet modeled)
options	lessonId, content	(not yet modeled)

This document was generated from verified Android source code, MongoDB Atlas collection screenshots, and inferred Node.js backend behavior. All field names, types, and values are sourced directly from the production codebase. — Charisma Editions · March 2026